ENS 210

31/12/2025

Osteogenesis Imperfecta Diagnosis Model

Presented By: Emir Ceylan

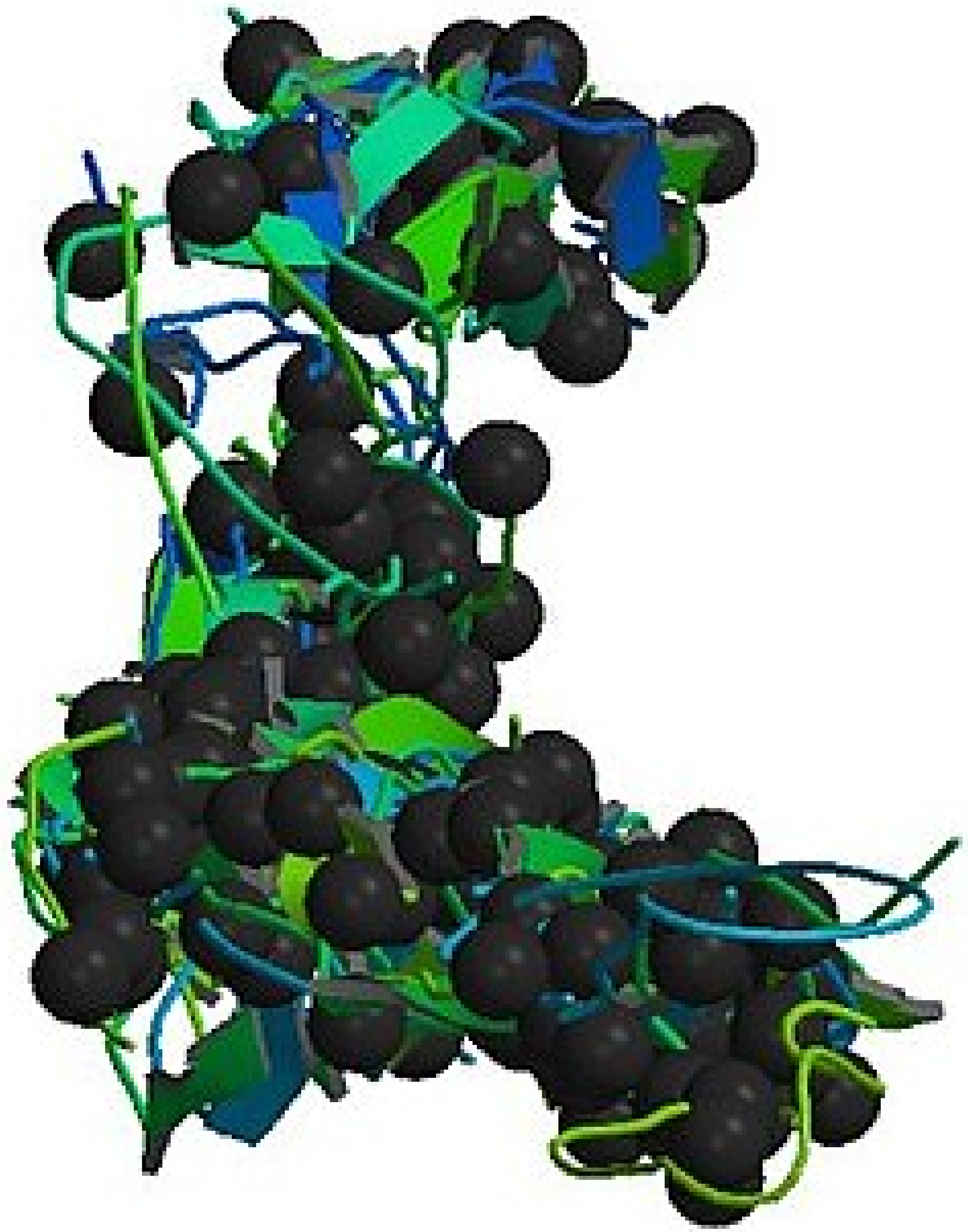
Osteogenesis Imperfecta

Osteogenesis Imperfecta (OI) is a genetic disorder characterized by bone fragility, primarily caused by mutations in the COL1A1 and COL1A2 genes encoding type I collagen.

These genes encode the α1(I) and α2(I) chains of type I collagen, the most abundant structural protein in bone, skin, tendons, and other connective tissues

Type I Collagen Structure

Heterotrimer composed of two α1(I) chains and one α2(I) chain
- Forms a characteristic triple helix structure
- Each chain contains a repetitive Gly-X-Y amino acid pattern
- Glycine (the smallest amino acid) must occupy every third position
- The tight packing of the triple helix requires glycine's small size at the core



[COLLAGEN, TYPE I, ALPHA 1]

DATA CONFIGURATION

Variant data were obtained from **ClinVar**

Exclusion Criterion:
- Variants of Uncertain Significance (VUS)
- Non-OI related variants
- Complex structural variants (>50 bp)

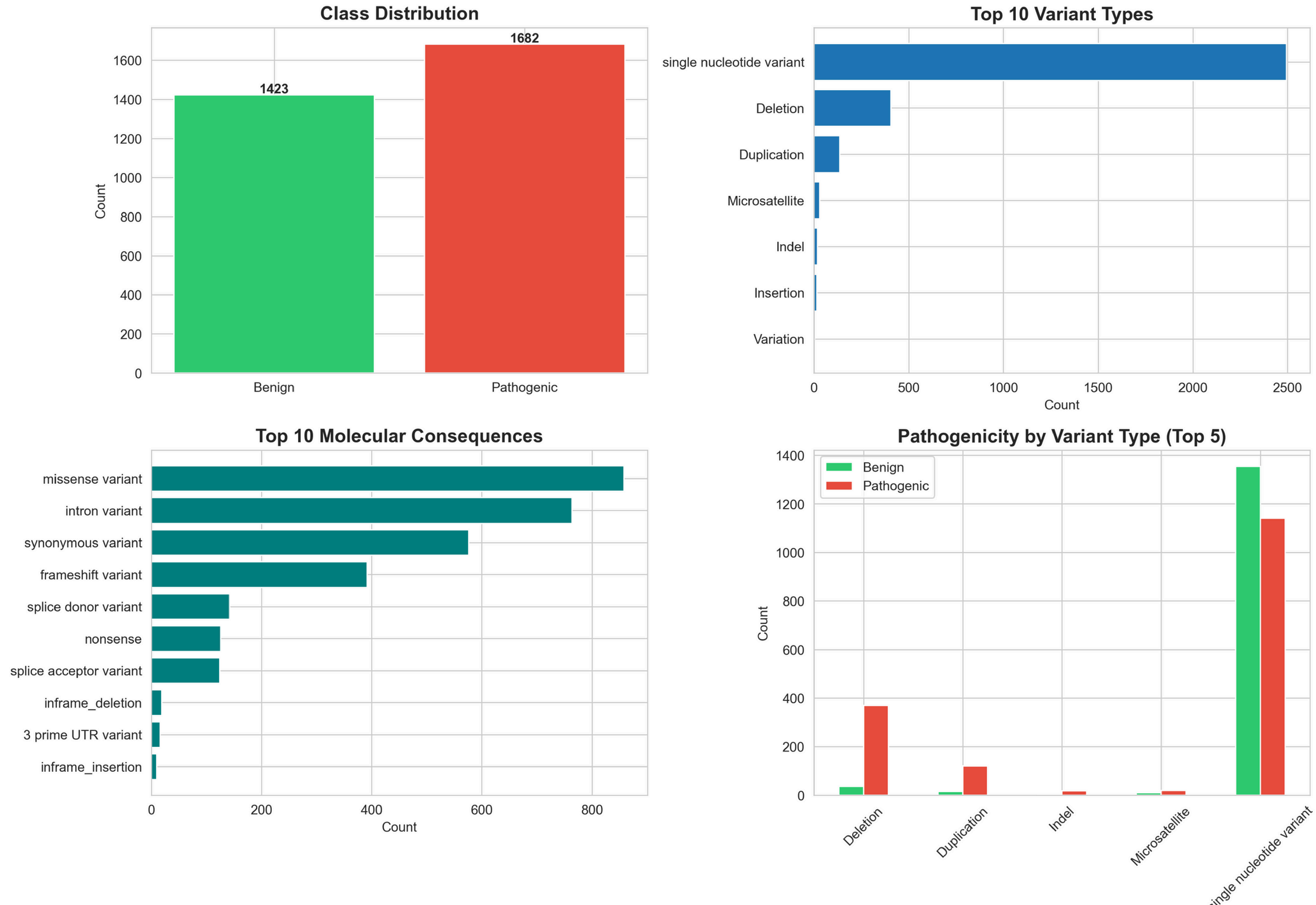
RAW VARIANTS: 3,584

CLEAN VARIANTS: 3,105

- COL1A1 1,739 variants (56.0%)
- COL1A2 1,366 variants (44.0%)

Pathogenic (label=1): 1,682 (54.2%)

Benign (label=0): 1,423 (45.8%)


Class Distribution
1423
1682
Count
0
200
400
600
800
1000
1200
1400
1600
Benign
Pathogenic
Top 10 Variant Types
single nucleotide variant
Deletion
Duplication
Microsatellite
Indel
Insertion
Variation
0
500
1000
1500
2000
2500
Count
Top 10 Molecular Consequences
missense variant
intron variant
synonymous variant
frameshift variant
splice donor variant
nonsense
splice acceptor variant
inframe_deletion
3 prime UTR variant
inframe_insertion
0
200
400
600
800
Count
Pathogenicity by Variant Type (Top 5)
Benign
Pathogenic
Count
0
200
400
600
800
1000
1200
1400
Deletion
Duplication
Indel
Microsatellite
single nucleotide variant

Method and Results

Accuracy = (TP + TN) / (TP + TN + FP + FN)

1. Logistic Regression

Baseline model; fast; interpretable coefficients

-Train Accuracy: 0.973

-Test Accuracy: 0.968

2. Random Forest

Handles non-linear relationships; provides feature importance; robust to overfitting

-Train Accuracy: 0.988

-Test Accuracy: 0.972

3. Support Vector Machine

Effective in high-dimensional spaces; memory efficient

-Train Accuracy: 0.977

-Test Accuracy: 0.967

4. Gradient Boosting

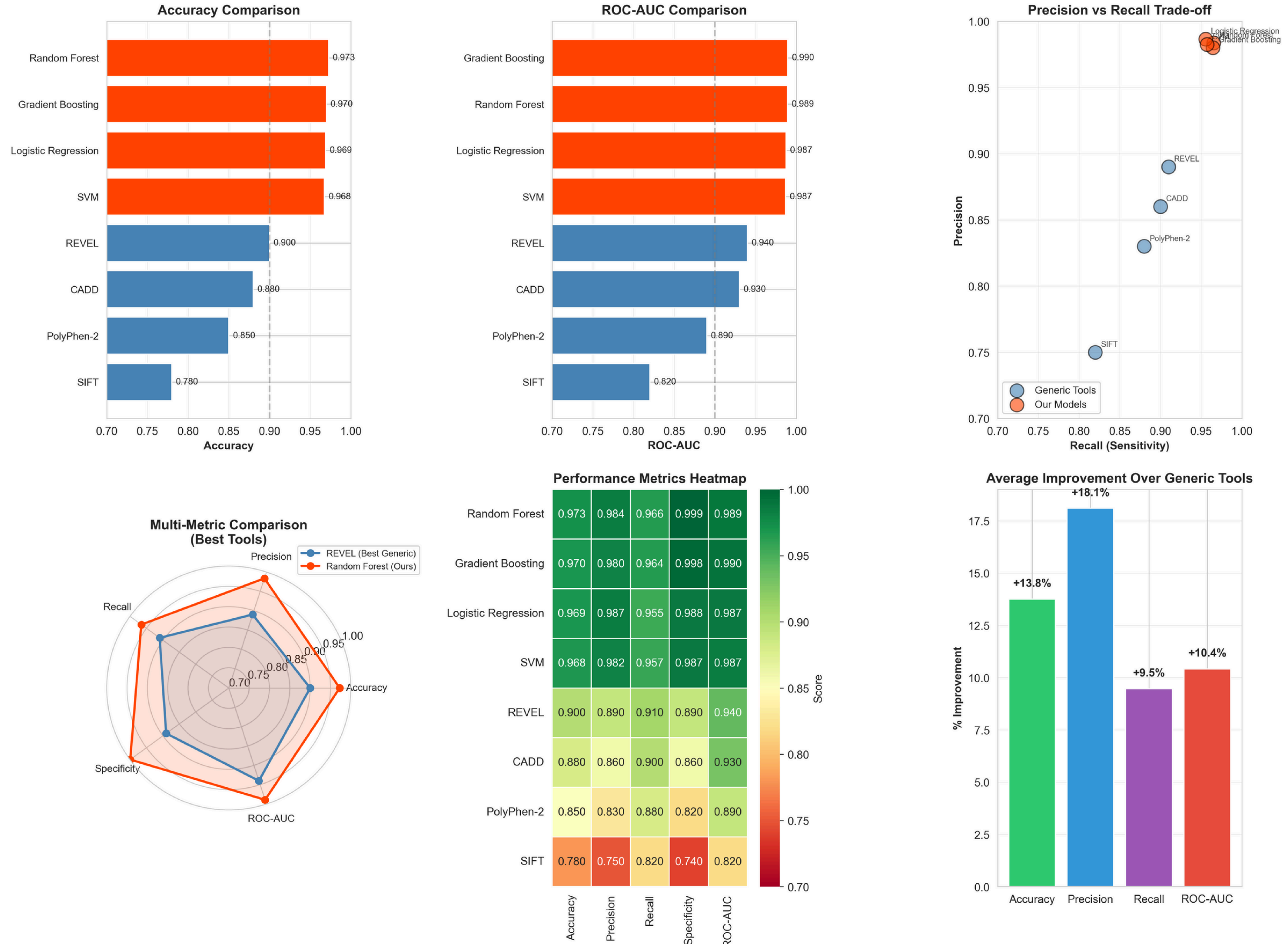
Often achieves best performance; builds trees to correct previous errors

-Train Accuracy: 0.989

-Test Accuracy: 0.970

Interpretation

- All models show <2% gap between training and test accuracy
- Minimal overfitting observed
- Hyperparameter choices (max_depth, regularization) were appropriate


Accuracy Comparison
Random Forest
Gradient Boosting
Logistic Regression
SVM
REVEL
CADD
PolyPhen-2
SIFT
0.973
0.970
0.969
0.968
0.900
0.880
0.850
0.780
0.70
0.75
0.80
0.85
0.90
0.95
1.00
Accuracy
ROC-AUC Comparison
Gradient Boosting
Random Forest
Logistic Regression
SVM
REVEL
CADD
PolyPhen-2
SIFT
0.990
0.989
0.987
0.987
0.940
0.930
0.890
0.820
ROC-AUC
Precision vs Recall Trade-off
Logistic Regression
SVM
Random Forest
Gradient Boosting
REVEL
CADD
PolyPhen-2
SIFT
Generic Tools
Our Models
Precision
Recall (Sensitivity)
Multi-Metric Comparison
(Best Tools)
REVEL (Best Generic)
Random Forest (Ours)
Precision
Recall
Accuracy
Specificity
ROC-AUC
0.70
0.75
0.80
0.85
0.90
0.95
1.00
Performance Metrics Heatmap
Random Forest 0.973 0.984 0.966 0.999 0.989
Gradient Boosting 0.970 0.980 0.964 0.998 0.990
Logistic Regression 0.969 0.987 0.955 0.988 0.987
SVM 0.968 0.982 0.957 0.987 0.987
REVEL 0.900 0.890 0.910 0.890 0.940
CADD 0.880 0.860 0.900 0.860 0.930
PolyPhen-2 0.850 0.830 0.880 0.820 0.890
SIFT 0.780 0.750 0.820 0.740 0.820
Accuracy
Precision
Recall
Specificity
ROC-AUC
Score
Average Improvement Over Generic Tools
+13.8%
+18.1%
+9.5%
+10.4%
% Improvement
Accuracy
Precision
Recall
ROC-AUC

Is this the Ultimate Tool?

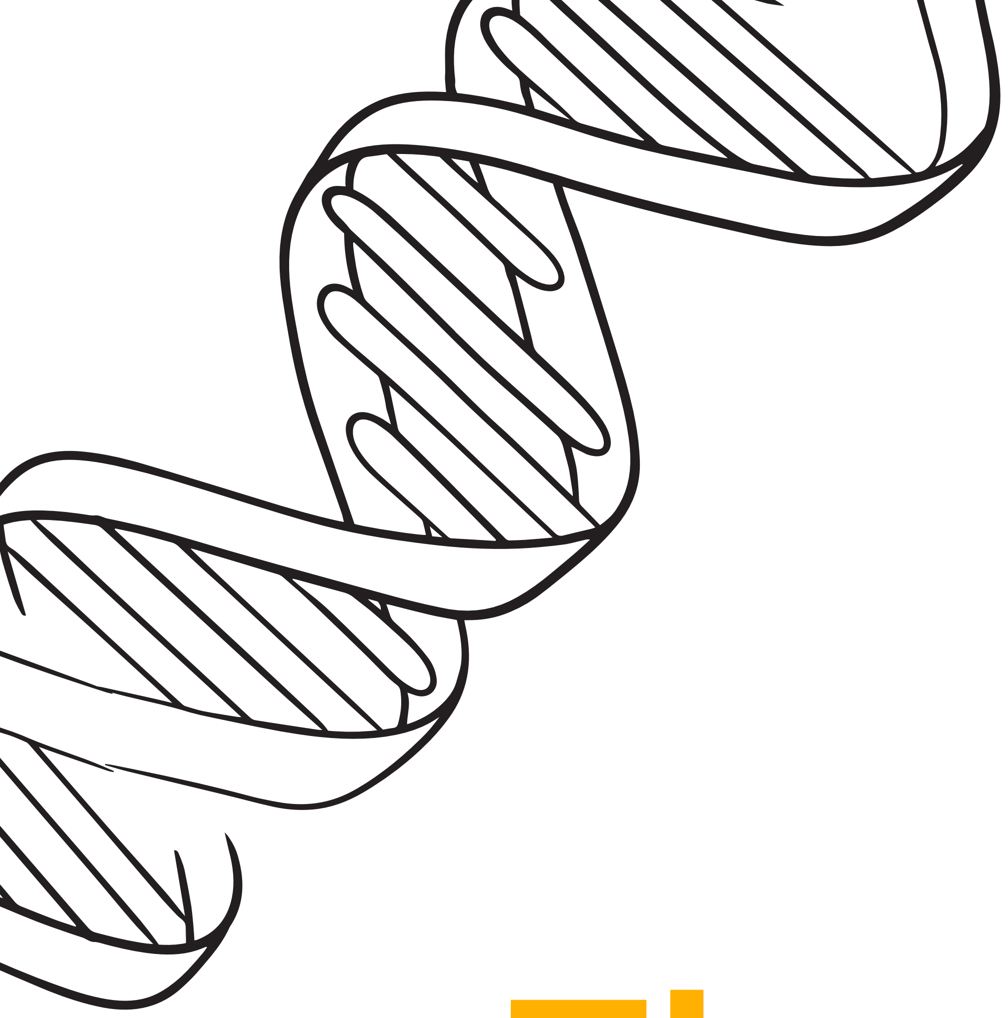
While the accuracy of my tool's ouputs are impressive (Over 7% improvement in each field compared to industry standard REVEL) we still can't say that it is the ultimate tool and there is no room for improvement

-Integrate 3D protein folding data (e.g., AlphaFold) to see how mutations physically bend the collagen.

-Expand to the other 15+ rarer genes known to cause OI.

Conclusion

Disease-specific approaches to diagnose are superior in diagnosing the patients. Merging the unique biological data we know about the disease with Machine Learning methods allow us to build Models that surpass generalized ones.



Thank You For Listening

Presented by Emir Ceylan